BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 239 · Thinking ahead: inputs, caching and reuse

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why omit cover colour from a simple book-loan model?

Q2. What is a reusable subroutine?

Q3. Calculate 7×7.

Q4. Why is caching squares safer than caching changing attendance data?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

input — Data passed into a computation.

output — Result of a computation.

precondition — Condition required before a process can run correctly.

cache hit — Requested value found in storage.

cache invalidation — Removing or refreshing outdated stored results.

New learning

Define the contract — The input is an integer, and the output is its square; overflow is a possible precondition concern.

Trace repeated work — First call for 5 calculates and caches 25; second returns the stored result.

Evaluate caching — Saves repeated calculation but costs memory. For changing data, cached values can become out of date.

Worked example

A cached ticket price for zone A is £3.40. The same zone is requested twice.

1. Predict first — The key A is already stored, so the lookup uses the cache.

2. Model and check — If pricing changes, the cached price must be updated or expired.

We Do · Practise with guidance

A cached attendance total is 12 but the live total changes to 15. How should the system respond?

```
cachedTotal=12; liveTotal=15;
```

Predict first

Using the old cached 12 gives a stale answer instead of the new total 15.

Model and check

Update or invalidate the cached value when attendance changes, or apply a suitable expiry policy.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For SquareCached(6), identify input, output and two preconditions or assumptions.

Question 2

Trace two calls of SquareCached(7) starting with an empty cache.

Question 3

Give three advantages/disadvantages of caching changing school data and a way to avoid stale values.

Exit Ticket · Answer before checking

Once a cached value is stored it is always correct.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Cover colour does not determine availability.

Q2. A function/procedure with a clear interface usable in multiple places.

Q3. 49.

Q4. Squares of the same integer do not change; attendance does.

You Do pages · Independent questions

Q1. [4] Input 6 [1]; output 36 [1]; valid integer input [1], no overflow for chosen integer type [1].

Q2. [6] First misses [1], computes 49 [1], stores 7→49 [1]; second hits [1], returns 49 [1] without recomputing [1].

Q3. [6] Faster repeated lookups [1], reduced repeated database work [1], uses memory [1], can return stale information [1], stale data can cause incorrect decisions [1], update/invalidate or expire on change [1].

Exit Ticket · Model answer

False. Dynamic data can change, leaving a stale result.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.